Optimizing Transport Predictive Modeling with Simulation-Based Statistical Inference

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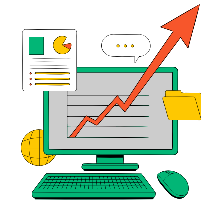


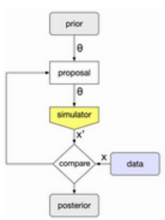
Introduction

Simulation-based statistical inference (SBI) uses computer simulations to help scientists understand and analyze complex data. It has many applications in:

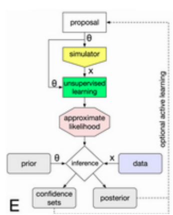
- Statistics
- Engineering
- Computer Science
- Robotics

Classical Methods

Approximate Bayesian Computation



Density Estimation



Research Focus

- Using SBI to analyze transport data
- Aiming to improve predictive modeling and decision-making

Why transportation and SBI?

- Analyzing transport data helps understand travel patterns, optimize infrastructure, and enhance network efficiency
- Transportation systems significantly impact economic development, environmental sustainability, and social equity
- Traditional statistical inference methods face challenges in capturing the complexity of transportation
- AI and machine learning improve inference quality, enabling us to handle more complex data

Data and Methodology

Data:

Table 1: Bureau of Transportation Statistics data on Indiana transportation

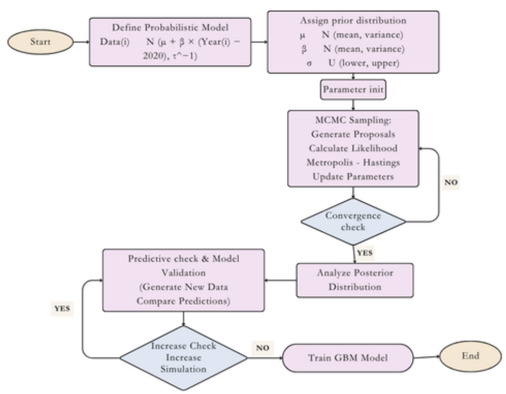
Year	Licensed drivers	Vehicles
2010	5,550,469	5,698,010
2011	5,669,665	6,132,770
2012	5,375,973	6,004,370
2013	4,500,403	5,574,030
2014	4,448,099	6,012,600
2015	4,467,848	6,045,114
2016	4,553,259	6,140,530
2017	4,553,584	6,170,034
2018	4,589,405	6,190,736
2019	4,589,405	6,223,460
2020	4,532,708	6,199,901
2021	4,636,114	6,241,291
2022	4,653,808	6,256,479

Techniques and Implementation

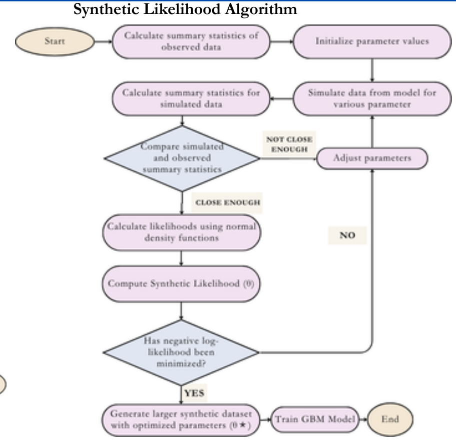
- Started using from 1 data point and gradually added more until 10 data points for predictions to check whether the method can be effective even when using minimal data
- Applied Approximate Bayesian Computation - Markov Chain Monte Carlo (ABC-MCMC) and Synthetic Likelihood methods to generate 100,000 total data points
- Used synthetic data to train Random Forest and Gradient Boosting Machine (GBM) models in R
- GBM showed better results, so focus only on the GBM model

Statistical Algorithms

ABC MCMC Algorithm



Synthetic Likelihood Algorithm



Results

Vehicles

Figure 1: Vehicles Predicted Values by Years of Data Used with Actual Data

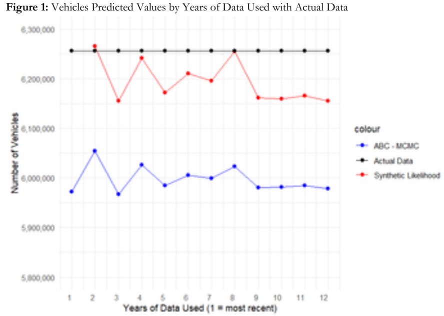
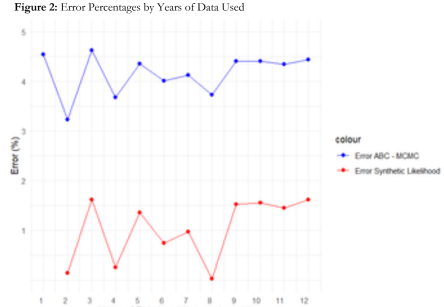


Figure 2: Error Percentages by Years of Data Used



Drivers

Figure 3: Drivers Predicted Values by Years of Data Used with Actual Data

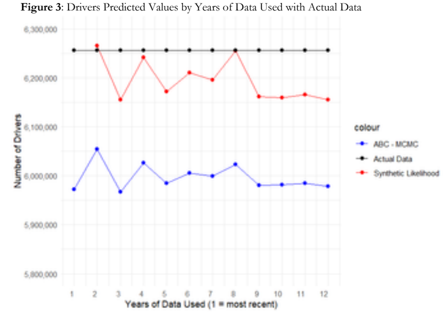
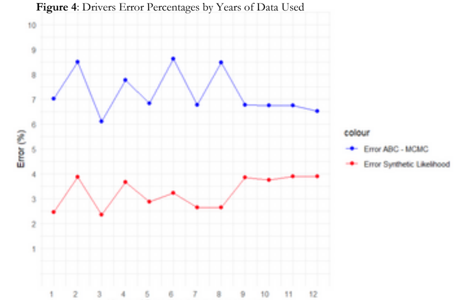


Figure 4: Drivers Error Percentages by Years of Data Used



Conclusion

Result

- Small difference between actual and predicted data
- Reliable with limited real-world data

Challenges

- High computational resource
- Dependence on model assumptions and data quality
- Time consuming

Impact on

- Optimize Investment
- Enhances network efficiency
- Promotes sustainable transportation solutions

Future Improvement

Computer Power

- Build models that require less computational power
- Implement efficient algorithms for data processing
- Optimize code to reduce runtime

Data

- Find better model assumptions and data quality
- Reduce margin of errors

References

[1] Cranmer, K., Brehmer, J., and Louppe, G., "The frontier of simulation-based inference," Proceedings of the National Academy of Sciences, vol. 117, no. 48, pp. 30055–30062, 2020.
[2] Frazier, D. T., Drovandi, C., and Nott, D. J., "Bayesian Synthetic Likelihood," arXiv preprint arXiv:2305.05120, 2023
[3] Marjoram, P., "Approximation Bayesian Computation," OA Genet., vol. 1, no. 3, 853, 2013, doi: 10.13172/2054-197x-1-1-853
[4] Manole, T., and Niles-Weed, J., "Statistical inference for optimal transport," The Annals of Applied Probability, vol. 34, no. 3, pp. 1108-1135, 2024
[5] Price, L. F., Drovandi, C. C., Lee, A., and Nott, D. J., "Bayesian synthetic likelihood," Journal of Computational and Graphical Statistics, vol. 27, no. 1, pp. 1–11, 2018
[6] Sadegh, M., and Vrugt, J. A., "Approximate Bayesian Computation using Markov Chain Monte Carlo simulation: DREAM(ABC)," Water Resources Research, vol. 50, pp. 6767–6787, 2014

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